

Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0-1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery, P=pretrain, ==tie): surgery 1 · pretrain 2 · tie 0 → PRETRAIN

	action_top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_nse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0-1)
ViT-g-2.0 pretrain_2X encoder	+11.62 [+9.260, +13.918]	+0.221 [+0.215, +0.226]		-0.138 [-0.142, -0.135]						0.67
ViT-g-2.0 pretrain encoder	+11.67 [+9.479, +13.918]	+0.105 [+0.102, +0.109]		-0.110 [-0.113, -0.107]						0.50
ViT-g-2.0 pretrain head	+0.603 [-1.589, +2.740]	+0.0018 [+0.002, +0.002]		-0.0003 [-0.001, +0.000]						0.04
ViT-g pretrain_2X encoder	+9.86 [+7.507, +12.154]	+0.037 [+0.035, +0.038]		+1.01 [+1.004, +1.009]						0.66
ViT-g pretrain encoder	+8.05 [+5.699, +10.411]	+0.034 [+0.033, +0.036]		+1.01 [+1.011, +1.016]						0.61
ViT-g pretrain head	+4.38 [+2.247, +6.575]	+0.0037 [+0.003, +0.004]		+1.01 [+1.006, +1.011]						0.45
ViT-g-2.0 surgery 3stage_DI encoder	+10.90 [+8.658, +13.206]	+0.113 [+0.110, +0.117]		-0.073 [-0.076, -0.070]						0.50
ViT-g-2.0 surgery noDI encoder	+11.29 [+8.986, +13.589]	+0.116 [+0.113, +0.120]		-0.070 [-0.072, -0.067]						0.52
ViT-g surgery 3stage_DI encoder	+9.42 [+7.178, +11.616]	+0.036 [+0.035, +0.038]		+1.02 [+1.014, +1.020]						0.65
ViT-g surgery 3stage_DI head	+8.11 [+5.973, +10.301]	+0.043 [+0.041, +0.044]		+1.01 [+1.011, +1.016]						0.62
ViT-g surgery noDI encoder	+8.27 [+6.027, +10.575]	+0.049 [+0.047, +0.051]		+1.01 [+1.007, +1.012]						0.63
ViT-g surgery noDI head	+8.77 [+6.575, +11.014]	+0.055 [+0.053, +0.057]		+1.01 [+1.010, +1.016]						0.66
WINNER	P	P	=	S	=	=	=	=	=	